

Sandeep Singh R

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PROFESSIONAL SUMMARY

PMP-certified Project Manager and XLRI-certified Product Manager with 10+ years of cross-functional experience spanning Embedded Systems, IoT, Semiconductor, and Automation domains. Proven track record in full project lifecycle delivery, Agile/Scrum execution, stakeholder management, and product roadmap ownership. Adept at leading engineering teams, managing budgets and risk, and driving on-time delivery with measurable quality outcomes. Equipped with a solid foundation in technical product thinking — including PRD/user-story definition and GTM alignment — making a natural fit for Junior Product Manager opportunities alongside senior Project Manager roles. Experienced in regulatory compliance (ISO, DO-178, MISRA C) and certified in both project and product disciplines.

CORE COMPETENCIES

Project Management	Agile & Scrum, PMP, PDLC, Program Management, Risk Register, Change Control, Milestone Tracking, Scope & Schedule Management, Stakeholder Management, Budgeting & P&L, Resource Allocation, Vendor Negotiation
Product Management	Product Roadmap, User Stories, PRD/MRD, Feature Prioritization, OKRs/KPIs, Go-to-Market (GTM), Customer Discovery, Voice of Customer (VoC), Competitive Analysis, Product-Market Fit
Aerospace Standards	DO-178C(Software),DO-254(Hardware, MISRA C
Process & Quality	Workflow Automation, QA & Validation, ISO / DO-178 / MISRA C Compliance, Design Reviews, Technical Documentation, CI/CD, Version Control
Tools & Platforms	JIRA, Bitbucket, Git, MATLAB GUI, Jython, Agile Boards, Sprint Planning
Leadership	Team Building, Cross-Functional Collaboration, Mentoring, Workshop Facilitation, Client Engagement

PROFESSIONAL EXPERIENCE

Project Manager | *Klug Avalon Mechatronics Pvt. Ltd.* Jun 2023 – Present
Domain: Safety-Critical Embedded Systems, IoT, Electro-Mechanical Products

- Led end-to-end project lifecycle for Safe Load Indicator (SLI) embedded systems — requirements definition, hardware/firmware design, system integration, qualification testing, and formal client sign-off — directly mirroring LRU development programme workflows for safety-critical applications.
- Owned PDR and CDR equivalent design review cadences, risk register management, and milestone scheduling across 5+ concurrent projects, achieving 100% on-time delivery.
- Reduced crane safety incidents by 40% through rigorous V&V processes, structured calibration protocols, and formal design reviews aligned to safety-critical standards.
- Managed P&L oversight including vendor negotiation, component sourcing, and procurement cost optimization — maintaining BOM compliance and budget targets.
- Drove cross-functional coordination across hardware design, firmware development, and QA validation teams — improving engineering workflow efficiency by 30% and reducing inter-team handoff delays.
- Achieved 95% client satisfaction through structured milestone reviews, clear deliverable sign-offs, and proactive stakeholder communication throughout the project lifecycle.
- Maintained change control discipline aligned to PMO standards; tracked and reported KPIs including cost, schedule, quality, and delivery milestones.

Assistant Manager – R&D | MPA Pvt. Ltd. May 2022 – Jun 2023

Domain: Embedded Product Development, Firmware Engineering

- Cut development cycles by 20% by implementing Agile practices, sprint retrospectives, and embedded system optimization techniques.
- Boosted device performance by 25% through targeted firmware enhancements and efficient hardware resource utilization.
- Maintained 95% client satisfaction by acting as the bridge between product and engineering teams — defining requirements, managing scope, and aligning delivery timelines.
- Improved vendor coordination and component sourcing efficiency by 15% through structured RFQ processes and supplier SLAs.

Senior Embedded Design Engineer | STPL

Domain: Embedded Hardware & Software Integration, System Validation

Mar 2021 – May 2022

- Drove hardware and software module design and integration for embedded products — managing PCB design validation, firmware integration, and structured design reviews consistent with CDR and integration test phases of LRU development.
- Improved system stability by 30% through systematic hardware/software integration supported by formal design reviews and structured test frameworks.
- Reduced product defects by 40% by implementing enhanced QA processes and validation frameworks consistent with DO-178/DO-254 V&V methodologies.
- Translated customer requirements into hardware/firmware feature specifications and delivery milestones — fulfilling the requirements-to-design traceability role essential in LRU programmes.

Career Break | Nov 2019 – Mar 2021

Personal bereavement (loss of father) coinciding with the COVID-19 pandemic. During this period, actively continued professional development as a freelance MATLAB and Embedded Systems Trainer/Developer — delivering online training sessions, developing client-specific MATLAB scripts and embedded prototypes.

Senior Software Developer (Client: Microchip Pvt. Ltd.) | Nexsemi Systems Pvt. Ltd. Oct 2018 – Nov 2019

Domain: Semiconductor Validation, MATLAB Tooling, Embedded Software

Validation Engineer (Mar 2019 – Sep 2019)

- Improved testing efficiency by 80% by automating manual test cases using Jython scripting, building reusable validation frameworks.
- Enhanced bug detection coverage by 25% through structured test planning; reduced regression testing effort by 15%.

MATLAB Developer (Oct 2018 – Mar 2019)

- Increased visualization efficiency by 35% by developing advanced MATLAB-based GUI dashboards used for engineering decision-making.
- Prototyped embedded algorithms enabling faster simulation, performance testing, and accelerated release cycles.

Core Embedded Contributions (Nexsemi)

- Boosted product efficiency by 20% via critical embedded software updates; reduced deployment errors by 25% through CI/CD automation and version control discipline.

Embedded Design Engineer | Monee & Company Pvt. Ltd.

Nov 2016 – Nov 2017

Company was subsequently shut down.

- Improved system stability by 20% designing and testing embedded hardware solutions; reduced prototyping timelines by 15% through PCB layout and firmware modularization.

Research Associate | PESIT (Higher Education Pursuit)

Oct 2012 – Aug 2014

Role concluded to pursue M.Tech (Signal Processing) full-time at SJCT, VTU.

- Enhanced data accuracy by 95% through research on acoustic systems and algorithm development.
- Published 5 National Conference Papers and 2 International Journal articles; mentored undergraduate research teams.

Design Engineer – Embedded | Apsis Solutions

Jun 2011 – Oct 2012

- Improved embedded performance by 15%; reduced hardware-software integration issues by 20% through systematic debugging and documentation.

KEY PROJECTS

Safe Load Indicator (SLI) System for Cranes

- End-to-end programme ownership: stakeholder requirements, HW/FW specification, system integration, qualification testing, and formal client sign-off — mirrors full LRU development lifecycle.
- Reduced safety incidents by 40%; reduced calibration setup time by 25% through enhanced calibration and structured V&V processes.

Spectrometer Design & Development

- Developed STM32 firmware improving data acquisition accuracy by 30%; optimised testing methodologies reducing hardware integration time by 20%.

Automatic Flight Control System (AFCS)

- Contributed to hardware rig design and FreeRTOS-based firmware integration for an aerospace control system; improved system responsiveness by 15% and reduced system failures by 25% through structured hardware and software testing.
- Managed cross-functional hardware design and firmware development coordination, qualification test activities, and formal design review artefacts — directly aligned to aerospace LRU programme practices.

MATLAB GUI Tooling (Microchip / Freelance)

- Delivered dashboards and simulation tools that accelerated engineering decision-making and training workflows.

EDUCATION & CERTIFICATIONS

M.Tech – Signal Processing | SJCIT, VTU | 2016

B.E – Electronics & Communication | SSEC, VTU | 2011

Certifications

- PMP® – Project Management Professional (PMI)
- Product Management Certification – XLRI
- Embedded Systems & Peripherals – Microchip India Pvt. Ltd.
- Online Certifications – Project Management & Embedded Systems (Coursera)

PATENT

Automobile Emergency Alerting Autonomous Device – Design Patent Filed, India (Reg. No. 453281-001)
| Mar 2025

ACHIEVEMENTS & ADDITIONAL

- Winner – Best Design, Embedded Design Contest (EFY)
- Published 5 National Conference Papers & 2 International Journals
- MATLAB Instructor – Udemy (Basic Course); conducted multiple engineering student workshops
- NCC 'B' Certificate Holder
- Passionate about adventure sports: rock climbing, rappelling, trekking